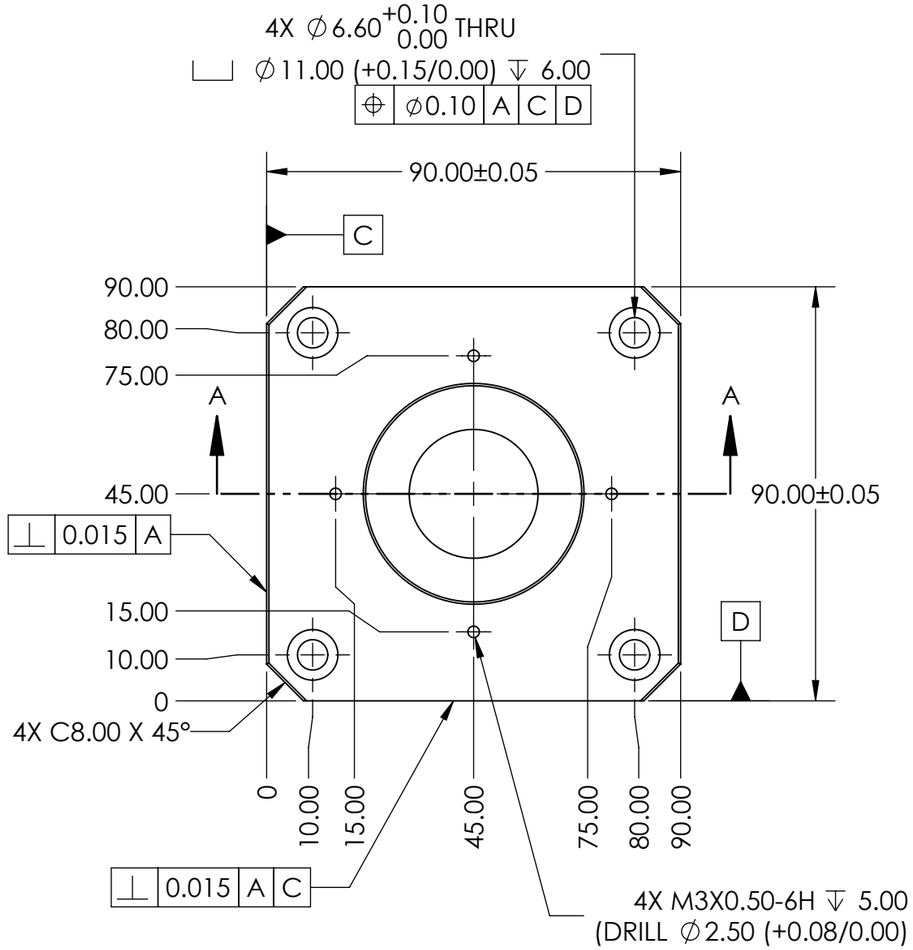
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

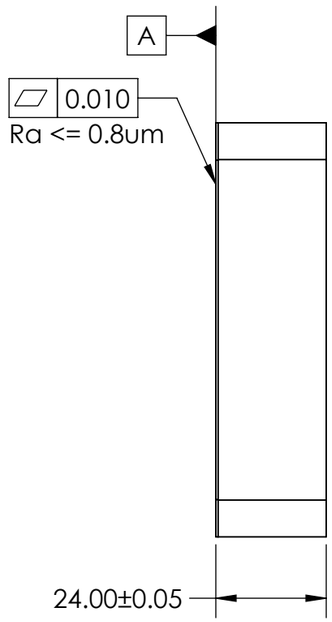
1. GENERAL & STANDARDS COMPLIANCE:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (W3-EV-MOTOR-ENDPLATE-4.2.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
   1.3 GENERAL TOLERANCES: LINEAR +/-0.05 mm, ANGULAR +/-0.2 DEG (PRECISION CNC TURN-MILLING).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
   1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

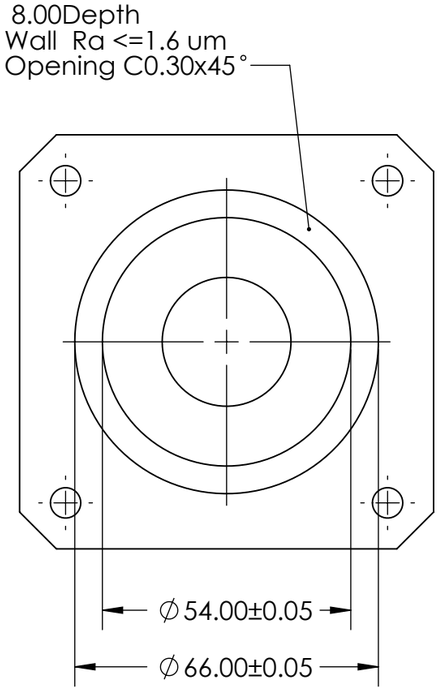
2. MATERIAL & EV AUTOMOTIVE ASSURANCE:
   2.1 MATERIAL OPTION A: FORGED AEROSPACE/EV ALUMINUM ALLOY 6082-T6 (EN AW-6082 / ASTM B221), HIGH THERMAL CONDUCTIVITY (k >= 170 W/mK).
   2.2 MATERIAL OPTION B: FORGED HIGH-STRENGTH ALUMINUM ALLOY 7075-T6 (AMS 4045).
   2.3 100% SUBTRACTIVE 5-AXIS CNC MACHINING ONLY. NO CASTING POROSITY OR ADDITIVE RESIDUALS PERMITTED (IATF 16949 COMPLIANT).

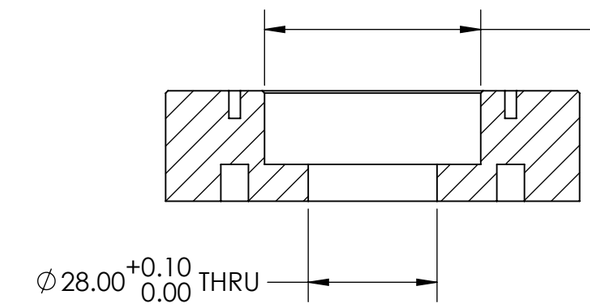
3. BEARING CHAMBER DYNAMICS & GD&T CONTROLS:
   3.1 PRIMARY DATUM [A] (STATOR MATING FACE): FLATNESS <= 0.010 mm, SURFACE FINISH Ra <= 0.8 MICRONS.
   3.2 SECONDARY DATUM [B] (BEARING BORE Ø47.000 +0.016/0.000 H7): CYLINDRICITY <= 0.005 mm, PERPENDICULAR TO [A] WITHIN 0.008 mm. SURFACE FINISH Ra <= 0.4 MICRONS TO ENSURE ZERO MICRON RUNOUT AT 20,000+ RPM.
   3.3 RESOLVER P.C.D. BOLT HOLES (4X M3): TRUE POSITION WITHIN Ø0.05 mm RELATIVE TO DATUM [B] AND [A] TO PREVENT ROTARY ENCODER ANGULAR TRACKING DRIFT.

4. THERMAL-HYDRAULIC & PRESSURE INTEGRITY:
   4.1 WORKING COOLANT: 50/50 WATER-ETHYLENE GLYCOL MIXTURE.
   4.2 COOLANT PASSAGE PROOF PRESSURE: WITHSTAND 0.6 MPa (6.0 BAR) HYDROSTATIC PROOF TEST FOR 5 MINUTES WITHOUT LEAKAGE (ZERO PRESSURE DROP).
   4.3 CORROSION MITIGATION: ANODIZE PER MIL-A-8625 TYPE III HARD COAT (20-25 MICRONS THICK) ACROSS COOLING JACKET FACES.





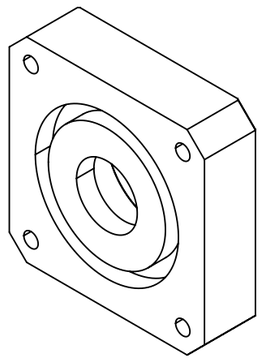




SECTION A-A

⌀47.000 +0.016 0.000 H7 ↧ 16.00
Opening C0.50 x 45°
Ra <= 0.4 um

⌭	0.005	
⊥	0.008	B
⏥	0.008	



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.05 mm, Angular ±0.2° (Precision 5-Axis CNC Turn-Mill)	FINISH: Hard Anodize Type III (MIL-A-8625, 20-25 µm thick, Anti-Glycol Corrosion)
DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING — REVISION: A

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE: 800V EV TRACTION MOTOR LIQUID-COOLED BEARING END PLATE

MATERIAL: Option A: Forged Aluminum Alloy 6082-T6 (EN AW-6082 / ASTM B221) / Option B: Al 7075-T6

DWG NO. W3-EV-MOTOR-ENDPLATE-4.2 — A3

WEIGHT: ~390.8 g (Al 6082-T6) / ~406.7 g (Al 7075-T6) — SCALE NTS — SHEET 1 OF 1